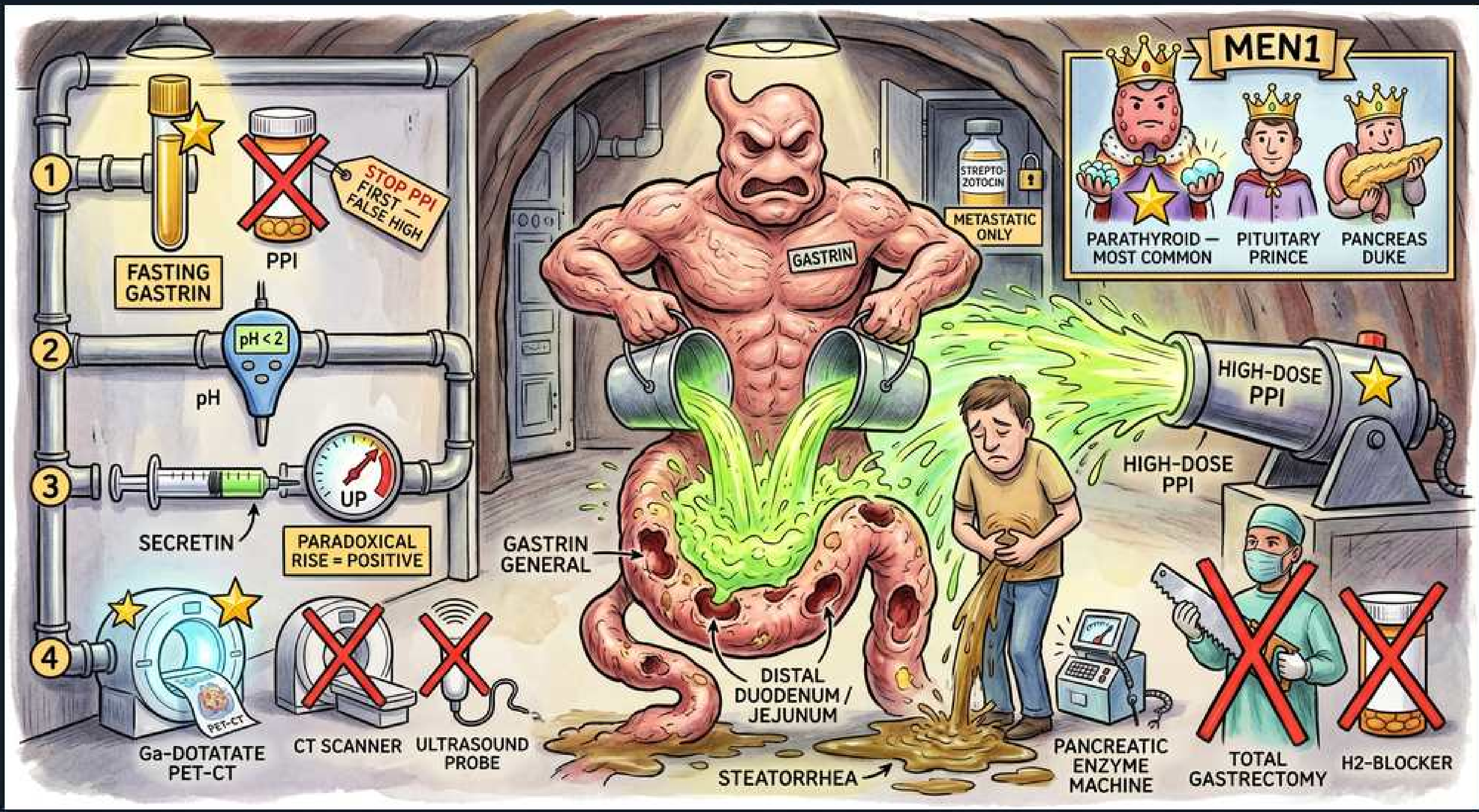
PUD — Zollinger-Ellison / Gastrinoma & MEN1



1. Gastrin General (the gastrinoma)

WHAT YOU SEE

A muscular figure made of stomach tissue labelled GASTRIN pouring buckets of green acid.

WHAT IT ENCODES

Gastrinoma = a gastrin-secreting neuroendocrine tumour (usually duodenum or pancreas, in the 'gastrinoma triangle'). Relentless hypergastrinaemia drives massive parietal-cell acid output → severe, multiple, recurrent ulcers. Suspect ZES with ulcers refractory to therapy, ulcers recurring after H. pylori eradication, ulcer + diarrhoea, ulcer + hypercalcaemia, or a strong family history.

BOARD TRAP

'Recurrent ulcer after successful H. pylori eradication = re-infection' → wrong. Re-think gastrinoma / ZES; check a fasting gastrin.

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2. Acid flood into duodenum

WHAT YOU SEE

Green acid gushing down into a swollen, ulcerated duodenum.

WHAT IT ENCODES

The huge acid load overwhelms duodenal buffering. Gastric pH stays inappropriately low (<2) even in the fasting state — this is what separates a true gastrinoma from other causes of high gastrin (which raise gastrin BUT with high/normal pH, e.g. atrophic gastritis, PPI use).

BOARD TRAP

'High gastrin alone confirms ZES' → wrong. You also need a LOW gastric pH (<2). High gastrin + high pH points to achlorhydria/atrophic gastritis or PPI effect.

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3. Ulcers in distal duodenum / jejunum

WHAT YOU SEE

Multiple ulcer craters extending unusually far down the small bowel, labelled DISTAL DUODENUM / JEJUNUM.

WHAT IT ENCODES

Ordinary peptic ulcers sit in the duodenal bulb or stomach. Ulcers in the SECOND/THIRD part of the duodenum or in the jejunum are a classic ZES fingerprint and a favourite board clue.

BOARD TRAP

'Multiple ulcers distal to the duodenal bulb are just severe H. pylori disease' → wrong; this distribution should trigger a gastrinoma work-up.

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4. Diarrhoea + steatorrhoea + broken enzyme machine

WHAT YOU SEE

A patient with watery diarrhoea and greasy STEATORRHOEA; a pancreatic enzyme machine snapped in half.

WHAT IT ENCODES

Acid floods the small bowel and INACTIVATES pancreatic lipase → maldigestion, steatorrhoea and watery (secretory) diarrhoea. The diarrhoea improves dramatically with a PPI — a useful clinical and exam clue that the cause is acid, not the pancreas itself.

BOARD TRAP

'Steatorrhoea in ZES means primary pancreatic insufficiency' → wrong; enzymes are made but acid-inactivated. The fix is acid suppression, not enzyme replacement.

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5. Step 1 — Fasting gastrin (gold star)

WHAT YOU SEE

A golden blood vial labelled FASTING GASTRIN, first in the pipeline, wearing a gold star.

WHAT IT ENCODES

First test = fasting serum gastrin. A markedly elevated level (often >10x normal, classically >1000 pg/mL) with gastric pH <2 is diagnostic. Modest elevations need the secretin test.

BOARD TRAP

'Order imaging first to find the tumour' → wrong; biochemical confirmation (gastrin + pH) comes BEFORE localisation.

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6. Red X on PPI — stop it first

WHAT YOU SEE

A PPI bottle with a big red X and a tag STOP PPI FIRST — FALSE HIGH.

WHAT IT ENCODES

PPIs themselves raise gastrin (loss of acid feedback) and cause FALSE-POSITIVE results. Stop the PPI (ideally ~1 week, with H2-blocker cover if needed) before measuring gastrin — this exact pitfall is heavily tested.

BOARD TRAP

'Measure gastrin while the patient is on their PPI' → wrong; the PPI falsely elevates gastrin. Stop it first.

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7. Step 2 — Gastric pH < 2

WHAT YOU SEE

A pH meter reading pH < 2.

WHAT IT ENCODES

Documenting acid hypersecretion (low pH) alongside high gastrin is what nails the diagnosis and excludes the benign causes of hypergastrinaemia.

BOARD TRAP

'Atrophic gastritis can give the same gastrin + pH picture' → wrong; atrophic gastritis gives HIGH gastrin with HIGH pH (achlorhydria).

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8. Step 3 — Secretin stimulation test

WHAT YOU SEE

A secretin syringe feeding a gauge whose needle jumps UP — PARADOXICAL RISE = POSITIVE.

WHAT IT ENCODES

If gastrin is only modestly elevated, give IV secretin: in ZES gastrin PARADOXICALLY RISES (normally secretin suppresses gastrin). A rise of ≥120 pg/mL is positive.

BOARD TRAP

'Secretin lowers gastrin, so a drop confirms ZES' → wrong; it's the paradoxical RISE that confirms gastrinoma.

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9. Step 4 — Ga-DOTATATE PET-CT (gold star)

WHAT YOU SEE

A glowing scanner printing a tumour map labelled Ga-DOTATATE PET-CT, with a gold star.

WHAT IT ENCODES

Best localisation = somatostatin-receptor imaging (Ga-DOTATATE PET-CT), the most sensitive modality; EUS is excellent for small duodenal/pancreatic primaries. Localisation only AFTER biochemical confirmation.

BOARD TRAP

'Plain CT is the most sensitive way to localise a gastrinoma' → wrong; somatostatin-receptor PET (Ga-DOTATATE) is far more sensitive, especially for small tumours.

10. Red X on CT scanner & ultrasound

WHAT YOU SEE

A CT scanner and an ultrasound probe each crossed out with a red X.

WHAT IT ENCODES

Conventional CT and transabdominal US miss small gastrinomas; they are not the answer for sensitive localisation.

BOARD TRAP

'Transabdominal ultrasound reliably finds gastrinomas' → wrong; small NETs are easily missed — use DOTATATE-PET / EUS.

11. High-dose PPI cannon (gold star)

WHAT YOU SEE

A cannon labelled HIGH-DOSE PPI with a gold star blasting and neutralising the acid flood.

WHAT IT ENCODES

Cornerstone medical management of ZES = HIGH-DOSE PPI to control acid hypersecretion (doses well above standard ulcer dosing). For sporadic, localised tumours → surgical resection can be curative.

BOARD TRAP

'H2-blockers are adequate first-line acid control in ZES' → wrong; the acid output is too high — high-dose PPI is required.

12. Red X on total gastrectomy & H2-blocker

WHAT YOU SEE

A surgeon with a TOTAL GASTRECTOMY saw and an H2-blocker bottle, both crossed out in red.

WHAT IT ENCODES

Total gastrectomy is historical/last-resort, not routine. H2-blockers under-treat the acid. Modern care = high-dose PPI ± tumour resection.

BOARD TRAP

'Total gastrectomy is the treatment of ZES' → wrong; high-dose PPI controls acid, and resection targets the tumour.

13. MEN1 crest — the 3 P's

WHAT YOU SEE

A royal crest reading MEN1 with three crowned figures: PARATHYROID (most common), PITUITARY, PANCREAS.

WHAT IT ENCODES

~20-25% of gastrinomas are part of MEN1 (autosomal dominant, MEN1 gene). The 3 P's: Parathyroid (primary hyperparathyroidism — the most common & usually first manifestation), Pituitary adenoma, Pancreatic/duodenal NETs (gastrinoma, insulinoma). MEN1 gastrinomas are often multiple → cure by surgery is unlikely, so they're managed medically.

BOARD TRAP

'Gastrinoma is the most common MEN1 manifestation' → wrong; primary HYPERPARATHYROIDISM is the most common and usually earliest. Always check calcium in a gastrinoma patient.

14. Streptozotocin locked away

WHAT YOU SEE

A chemo bottle labelled STREPTOZOTOCIN in a cabinet marked METASTATIC ONLY.

WHAT IT ENCODES

Streptozotocin-based chemotherapy is reserved for advanced/metastatic gastrinoma, not a first-line step in the typical work-up question.

BOARD TRAP

'Start streptozotocin chemo for a newly found localised gastrinoma' → wrong; resect localised disease; chemo is for metastatic NET.